

Conductors at RF and Transmission Lines

Week 1

At DC

Resistance = Length / (Conductivity * Cross Sectional Area)

Resistance = Length * Resistivity / CSA

At DC, whole conductor is used to carry the current

At AC

Resistance increases with frequency because of skin effect (inductance)

$$X_L = j\omega L \text{ where } \omega = 2\pi f \text{ so } X_L = j 2 \pi f L$$

RF current flow migrate to the outer edge of the conductor to flow along path with least resistance

Skin Depth Formula

Skin Depth Formula

$$\delta_s = \sqrt{\left(\frac{1}{\pi f \mu \sigma}\right)} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{\rho}{\pi f \mu}}$$

$$\omega = 2\pi f$$

Where f is the operating frequency (Hz)

μ is the permeability of the conductor (H/m)

$$\mu = \mu_r \mu_0 \quad (\mu_r \text{ is usually } 1 \text{ and } \mu_0 = 4\pi \times 10^{-7})$$

σ is the conductivity of the conductor (S/m)

ρ is resistivity ($\Omega \cdot \text{m}$)

Common Transmission Lines

- TV and RF cables
- Microstrip
- Twisted Pair

Size of Transmission Lines

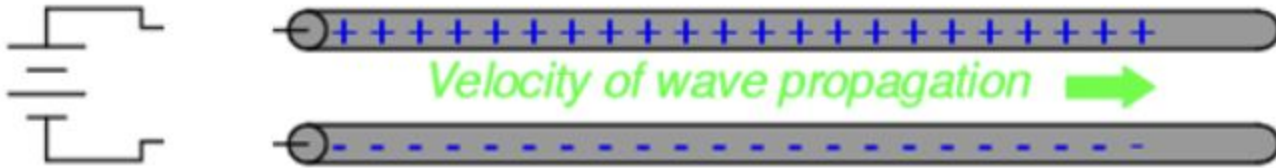
Length of cable $> 0.1 \lambda$ (1/10 of wavelength of transmitted signal)

If not:

- Reflection
- Radiation

Characteristic Impedance

A transmission line of infinite length, what happens when a battery is connected?



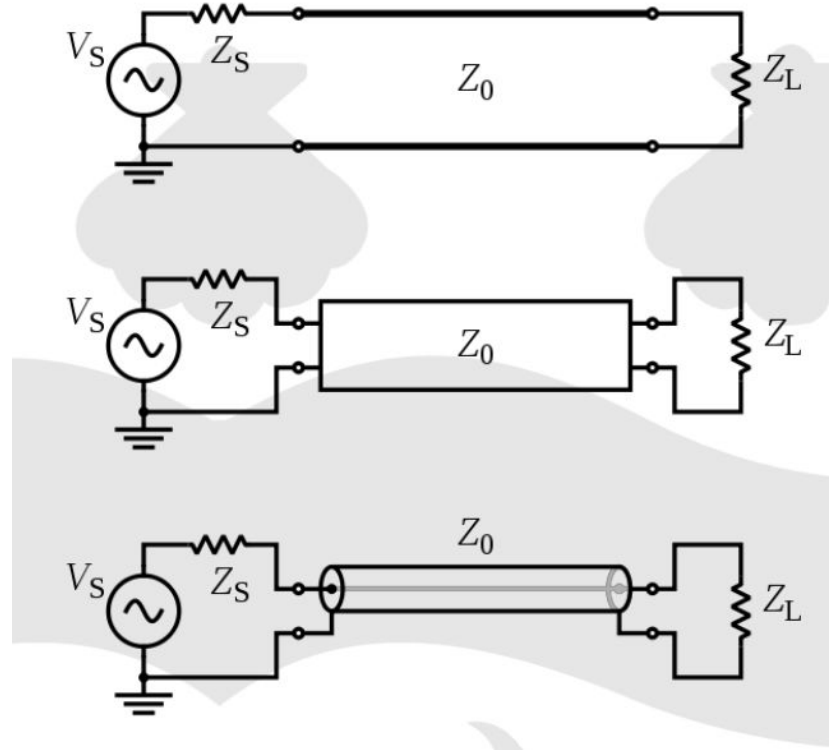
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Transmission Line Models

Z_s = input impedance

Z_0 = characteristic impedance

Z_L = load impedance



Reflection Coefficient, $\Gamma = \frac{Z_{IN} - Z_0}{Z_{IN} + Z_0} = \frac{Z_L - Z_0}{Z_L + Z_0}$

dBs and Link Budgets

Week 2

Why dB?

dBs compress the dynamic range and put everything on the same scale

For example:

Distance	dB
1mm	-30dB
1m	0dB
10m	10dB
100m	20dB
1km	30dB
10km	40dB

Transmit & Receive Power

Communication Systems	Received Power	Transmit Power
GSM	-51dBm	Varies @ 900 & 1800MHz
GPS	-155 dBW (-125dBm)	25.6W @ 1573MHz
WiFi	-75 dBm or better	100mW (1W) @ 2.4GHz
Voyager Space Ships	-231dBm	18W @ 8GHz

Thermal Noise

$$N = -204 + 10 \log_{10}(B) \quad (dBW)$$

$$N = -174 + 10 \log_{10}(B) \quad (dBm)$$

Noise Factor & Figure

$$\text{noise factor } F = \frac{\left(\frac{S}{N}\right)_{in}}{\left(\frac{S}{N}\right)_{out}}$$

$$\text{or in dB, noise figure } F(\text{dB}) = \left(\frac{S}{N}\right)_{in} (\text{dB}) - \left(\frac{S}{N}\right)_{out} (\text{dB})$$

$$N = -204 + F(\text{dB}) + 10\log_{10}(B) \quad (\text{dBW})$$

Path Loss

R is distance from transmitter

Lambda is c/f

$$L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

L Matching and Smith Charts

Week 3

	Symbol	Component Value	Impedance at 1kHz	Impedance at 1MHz		Admittance at 1MHz (1/Z)
$Z_R = R$		10 Ω	10 Ω	10 Ω	$Y = 1/R$	0.1S
$Z_L = j\omega L$		10mH	$j62\Omega$	$j62k\Omega$	$Y = 1/j\omega L$	$-j16\mu S$
$Z_C = 1/j\omega C$		10nF	$-j16k\Omega$	$-j15 \Omega$	$Y = j\omega C$	0.066S

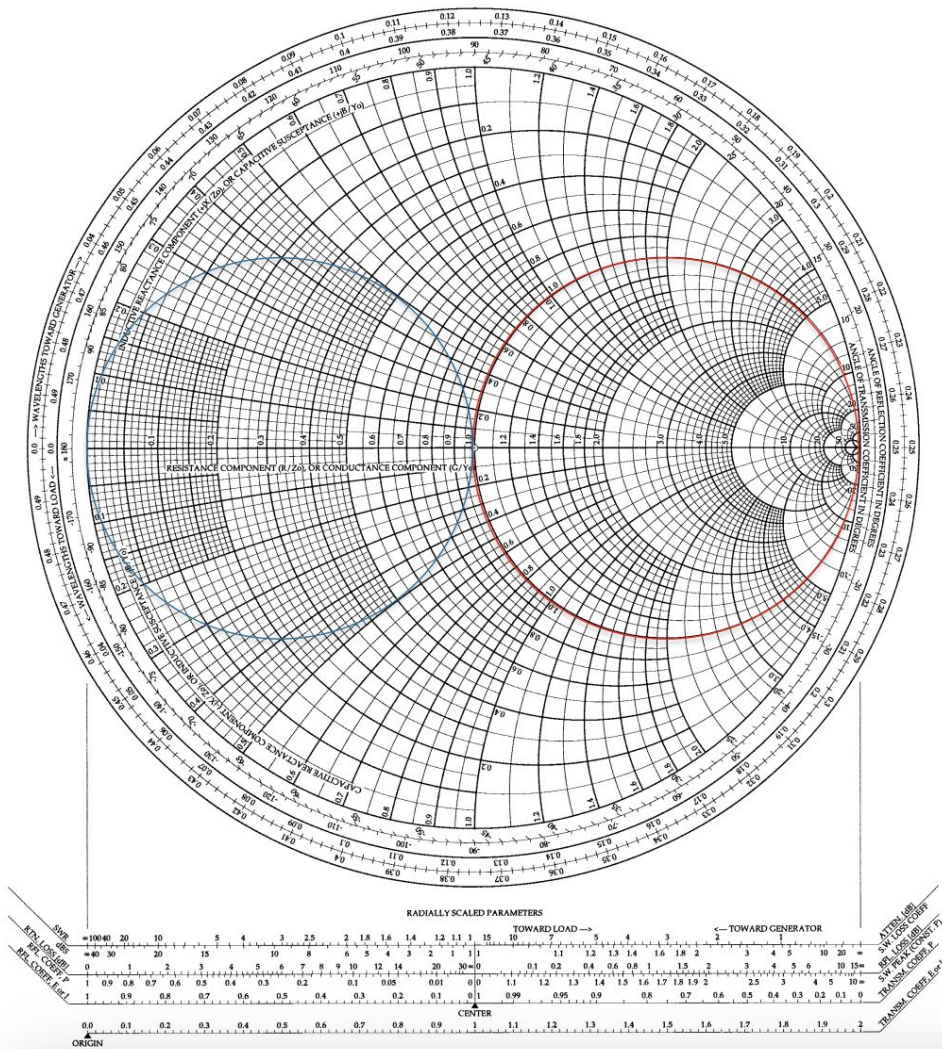
Where $\omega = 2 \pi f$ and f = frequency.

Inductance and Capacitance

+j (up the smith chart)

-j (down the smith chart)

	Series	Shunt (Parallel)
	-j	+ j
	+ j	-j



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